

Conflict on the campaign trail? How campaign effort and electoral competitiveness shape affective polarization

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ABSTRACT

Concerns about the consequences of affective polarization for governing and democracy have resulted in a strong surge of publications on the topic. This literature has already provided much insight into the individual-level correlates of affective polarization, but we know much less about how the political context in which voters choose can contribute to affective polarization. We theorize that the intensity of election campaigns increases affective polarization, which we capture through indicators of (a) campaign spending, from both in-parties and out-parties, and (b) electoral competitiveness. Using the British case and leveraging within-country and over-time variation in these indicators, we find support for our expectations. Specifically, we show first that in-party spending is positively associated with affective polarization, leading both to increased in-party like and out-party dislike. Second, we show that competitiveness is positively associated with out-party dislike—individuals who experience more competitive electoral contests grow to dislike parties that they compete with to a greater degree. We contend that this mechanism could contribute to the within-country variations in affective polarization found in previous research.

1. Introduction

Concern is growing over the strong political divisions and partisan conflict that characterize electorates in much of the democratic world. Partisan disagreement over the world's most pressing problems appears entrenched (Falkenberg et al., 2022; Rodriguez et al., 2022) and citizens seemingly display a concerning willingness to prioritize tribal loyalties over democratic norms (Gidengil et al., 2022; Graham and Svobik, 2020). One salient aspect of this divide is affective polarization, or the tendency for partisans to exhibit dislike for opposing parties and their supporters and display ingroup biases in evaluations of and behavior toward in-party and out-party supporters (Iyengar et al., 2012; Iyengar and Westwood, 2015). The literature devoted to affective polarization—its prevalence, causes and consequences—is expansive (e.g. Boxell et al., 2024; Harteveld, 2021; Hobolt et al., 2021; Webster and Abramowitz, 2017). As part of this research effort, comparative studies have examined cross-national variation in affective polarization (Reiljan, 2020; Wagner, 2021). However, much less focus has been devoted below the national level. Whilst initial evidence establishes the existence of substantial within-country heterogeneity in affective polarization (Bettarelli

et al., 2023), potential factors explaining this variation have yet to be assessed.

This paper proposes and tests an explanation for these within-country differences in affective polarization, examining the potential role of election campaigns and heterogeneous campaign effects as a source of local variation in inter-partisan hostility. We focus on two specific mechanisms, analyzing whether spatial variation in the campaign efforts of political parties during elections or in the competitiveness of district contests (i.e., the marginality of specific races) prompts differing degrees of polarization among voters. Campaign efforts and competitiveness, we argue, likely contribute to affective polarization among voters over the course of election campaigns by increasing partisan identity salience, prompting greater cognitive engagement and amplifying out-group threat perceptions.

We test this proposition using British Election Study Internet Panel (BESIP) data from the 2015, 2017 and 2019 general elections in Britain (Fieldhouse et al., 2024). Britain makes for a strong case in which to examine the impact of heterogeneous campaign effects on affective polarization for two reasons. First, its single-member district plurality

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system produces district-level variation in campaign effort and competitiveness which we can exploit to test our hypotheses. Second, the rare occurrence (for Britain) of three elections over a 6-year period and the British Election Study team's efforts to collect panel data throughout this time period allow us to study the role of campaigns under varying electoral conditions—enhancing the robustness of our findings.

Our results provide evidence to suggest that both campaign effort—which we operationalize through party spending—and competitiveness contribute to affective polarization. Individuals who reside in parliamentary constituencies in which their in-party expend more campaign funds grow more affectively polarized over the course of the campaign, both via increased in-party like and greater out-party dislike. Alongside this, individuals who experience marginal constituency races grow to dislike the parties that they compete with more than those individuals who reside in non-competitive constituencies.

Our paper makes two important contributions to the literature. First, our analyses shed light on the sources of within-country variation in affective polarization, by highlighting the role of the local electoral context. Second, we add to the literature that has shown that elections increase affective polarization by theorizing and testing the role of different aspects of an electoral context—party spending and competitiveness—on affective polarization.

Returning to the central question of within-country variation in affective polarization, we argue that in single member district plurality systems such as the UK, the spatial distribution of campaign characteristics offers a potential explanation. Whilst the dynamics of party competition vary from one election to the next as the popularity of extant parties shift and new parties emerge, there is also an aspect of stability to the spatial distribution of competition. In other words, some districts (constituencies in the language of British politics) are always more competitive than others. Thus, over time, these differences in competitiveness could produce local variation in affective polarization. In offering initial evidence of a mechanism by which regional variation in affective polarization emerges, our findings represent an important step forward in understanding the distribution of inter-partisan hostility within countries. In doing so we also speak more generally to the literature on elections and affective polarization, providing novel evidence as to the role of election characteristics in facilitating partisan hostility.

2. Theory

Comparative research has established the existence of affective polarization across countries and revealed substantial cross-national variation in levels of affective polarization (Garzia et al., 2023). Countries differ both in the extent to which voters are polarized (Gidron et al., 2023; Reiljan, 2020; Reiljan et al., 2024) and in their overtime trends, with affective polarization growing in the US faster than in other countries (Boxell et al., 2024). However, this scholarship has predominantly focused on differences *between* countries (e.g. Reiljan, 2020; Wagner, 2021), and differences *within* countries have not been subjected to the same level of analysis.

There is a scarcity of research that examines variation in affective polarization below the national level. One notable exception is the work of Bettarelli et al. (2023), who use data from European elections included in the Comparative Study of Electoral Systems (CSES) project to show that within-country variation in affective polarization is as prominent as the differences that exist between countries. They also present evidence that this sub-national heterogeneity is growing over time. These findings demonstrate that regional variation in affective polarization is substantively important, but we know little about what might be driving this heterogeneity. In related work, Bettarelli and Van Haute (2022) connect sub-national differences in affective polarization to regional economic inequalities. Aside from this single paper, however, we still have limited insights into the causes of within-country variation in affective polarization.

Here, we draw attention to the impact of election campaigns in generating local heterogeneity in affective polarization. A nascent literature has established that elections drive affective polarization among voters (Hernández et al., 2021). As the theory goes, elections raise the salience of politics and in doing so activate partisan identities among the electorate (Singh and Thornton, 2019). This activation of partisanship then, fosters affective polarization. Specifically, partisanship represents an emotional and psychologically meaningful attachment to a political party and its supporters (Huddy et al., 2015), and in order to maintain a sense of self-esteem, in-group cohesion and positive distinctiveness from out-groups, individuals engage in a process of in-group bias, favourably evaluating in-groups relative to out-groups (Tajfel and Turner, 1979; Turner et al., 1987). The proclivity toward in-group bias is conditioned by context—as identities become more salient, the tendency to display in-group bias grows (Mullen et al., 1992; Oakes, 1987). Work that has argued that elections generate affective polarization points out that elections highlight and amplify political conflict and party competition. Elections also attract large amounts of media attention to the party political realm, provide a platform for voter engagement and offer tangible outcomes to the electorate, all of which serve to raise the salience of partisan identity (Singh and Thornton, 2019). Via this mechanism, elections heighten in-group biases and thus lead to (greater levels of) affective polarization (Hernández et al., 2021; Bassan-Nygate and Weiss, 2022; Singh and Thornton, 2024).

Extant research has so far operationalized the salience of elections using the temporal proximity to election day. Using variation in the date of interviews among survey respondents (assumed to be random), a number of studies have shown that survey respondents interviewed closer to election day are more affectively polarized (Bassan-Nygate and Weiss, 2022; Hernández et al., 2021). The findings of these studies suggest that elections and political contestation between parties indeed drive affective polarization. Additional scholarship has explored heterogeneity in this effect. Evidence points to a number of sources of potential variation: individual differences, such as perceptions of ideological differences between parties, strength of party identification or political engagement (Hernández et al., 2021; Hahm et al., 2023); country-level factors such as elite polarization (Gidron et al., 2023; Hahm et al., 2023); and electoral outcomes (Gidron and Sheffer, 2024; Sheffer, 2020).

However, heterogeneity in the nature of campaigns themselves and how this might prompt variation in affective polarization has received comparatively less attention (for notable exceptions, see Fasching et al., 2024; Iyengar et al., 2012). Political parties operate with finite resources and thus make active decisions about where those resources are distributed during election campaigns. As a result, the campaign effort expended by parties—defined as ‘the combination of activities aimed at maximising electoral gains’ implemented by parties and candidates (Trumm et al., 2017, p.462)—varies spatially within each campaign. This is particularly the case in single-member district plurality systems (SMDP). In elections held under SMDP, the chance of success for each party varies from seat to seat. Parties then undertake a form of cost-benefit analysis to determine how best to distribute their campaign efforts, expending greater campaign effort in seats in which success is a possibility but likely to be met with competition from other parties (Cutts and Johnston, 2015; Fisher et al., 2016; Karp et al., 2008). In addition, there is much heterogeneity in the nature and experience of an election for voters in single-member district plurality systems. The regional distribution of political support for parties produces safe seats (in which one party is assumed to win with ease) and tight marginals (uncertain contests in which two or more parties possess a reasonable chance of winning the seat).

This means that political campaigns within single-member district plurality elections—and the experience of these campaigns for voters—are heterogeneous. This raises the possibility that the impact of political campaigns on affective polarization could also exhibit spatial variation. Specifically, we posit that (1) voters who reside in districts in which parties expend greater campaign efforts and (2) voters who reside in

competitive districts will become more affectively polarized over the course of an election campaign. We discuss the reasoning for each of these two expectations in turn.

2.1. Party campaign effort

The first campaign characteristic that we anticipate could produce local variations in affective polarization is campaign effort. However, we anticipate that the individual-level consequences of campaign efforts for affective polarization are likely influenced by whether the campaigning is from an individual's in-party, or from out-parties. Campaign efforts (such as the provision of party leaflets, targeted online advertising or door-to-door canvassing) from an individual's in-party serve as cues for their party identity, and encountering such cues both raises the salience of the relevant identity and increases the likelihood of in-group identification (Mackie et al., 2008). Given evidence that the more salient an identity, the more likely one is to think as an in-group member and exhibit in-group bias (Mullen et al., 1992; Oakes, 1987), we thus anticipate that in-party campaign effort will lead to greater levels of affective polarization.

A second mechanism by which in-party campaign effort could boost affective polarization is via the provision of information. In addition to influencing the salience of group identities, campaign effort increases the availability of information about parties and policies and reduces information acquisition costs for voters (Trumm et al., 2017; Chapman and Palda, 1983). This information is likely to be partial—campaign information from a specific party seeks to paint that party in a favourable light and offer criticisms of competitors. Importantly, as this information comes from the individual's favoured party, it is likely to be accepted: experimental evidence demonstrates that in-party cues and information shape voter preferences and perceptions (Bisgaard and Slothuus, 2018; Slothuus and de Vreese, 2010; Slothuus and Bisgaard, 2021). Thus, a voter in a seat targeted by greater levels of in-party campaign effort will be exposed to a greater quantity of in-party campaign information, which is likely to lead to more favourable in-party perceptions and more negative out-party perceptions. Given this, we anticipate that:

H1: In-party campaign effort will be positively associated with affective polarization

In contrast, campaign effort from out-parties potentially has different consequences for affective polarization. Campaign efforts from out-parties—such as advertising, leaflets or door-to-door canvassing from out-party volunteers—will constitute attitudinally incongruent information for a voter, by criticizing in-party candidates and elites or offering alternative perspectives on salient issues. We see two possible ways in which voters could process such counter-attitudinal information. The first is that counter-attitudinal campaign efforts prompt more systematic, open-minded information processing and decision-making (Matthes, 2012), resulting in attitudinal updating, more balanced judgments of parties and candidates, and reduced levels of affective polarization. Evidence for this proposition comes from two sources. First, research shows that campaign effort prompts open-minded thinking and less polarized candidate evaluations among voters (Kam, 2006); second, research indicates that exposure to counter-attitudinal information can delay voting decisions by prompting uncertainty and more effortful decision-making (Matthes, 2012; Nir and Druckman, 2008).

The second possibility is that voters interpret counter-attitudinal campaign information from out-parties via processes of partisan motivated reasoning (Kunda, 1990; Petersen et al., 2013). Experimental evidence across national contexts indicates that individuals are more likely to reject or disagree with political arguments and issue frames when those arguments come from out-parties (Bolsen et al., 2014; Druckman et al., 2013; Slothuus and de Vreese, 2010). Additionally, research suggests that information consumption among partisans is selective, with more attention paid to attitudinally congruent information (Iyengar and Hahn, 2009), even during elections (Peterson et al., 2021).

Partisanship thus may act as a filter which prompts individuals to reject or ignore counter-attitudinal information from out parties, having null effects on affective polarization. Given these conflicting possibilities, we leave our expectations for out-party campaign effort as an open question:

RQ1: What are the consequences of out-party campaign effort for affective polarization?

2.2. Electoral competitiveness

The second campaign characteristic with potential implications for affective polarization is constituency-level competitiveness. In competitive local contests in which the margin of victory is likely to be small, voters have greater reason to believe that their vote has an impact (Downs, 1957; Riker and Ordeshook, 1968). Perceptions of decisiveness influence voter engagement—the closer the race and the more pivotal a voter feels, the more likely they are to engage with that election (Duffy and Tavits, 2008; Levine and Palfrey, 2007). In addition, both observational and experimental evidence suggests that closely-fought contests (or expectations of closeness) prompt heightened cognitive engagement from voters, who are more likely to seek out and retain information about the election (Kam and Utych, 2011; Roy et al., 2015). As with campaign efforts, voter engagement and information seeking increases the likelihood that voters encounter partisan identity cues, priming partisanship, and hence leading to in-group bias and affective polarization.

A second mechanism by which we expect competitiveness to increase affective polarization is social comparison and heightened out-group threat. During elections, opposition parties constitute a threat to an individual's political in-group: both as a material threat to the in-group's objective of electoral success, and a symbolic threat to group status (Huddy et al., 2015). Research from social psychology shows that perceived threat from out-groups leads to in-group bias and denigration of those groups (Shepherd et al., 2018; Smurda et al., 2006), meaning that elections likely prompt out-party hostility.

Importantly, we expect this relationship to be greater in competitive contests. In situations of intergroup competition—such as an election—closer contests both amplify the degree of social comparison between in-group and out-group that individuals make (Garcia et al., 2013) and increase individual desire to 'beat the other side' (Malhotra, 2010). Extending this logic to elections, Miller and Conover (2015) argue that when elections are more competitive, voters perceive the out-group as more threatening. Supporting this proposition, the authors find that individuals living in more competitive districts express greater anger and hostility toward out-party supporters, using data from a US election (Miller and Conover, 2015). Consequently, we anticipate that:

H2: Constituency-level competitiveness will be positively associated with affective polarization

It is important to note that theoretical approaches to competitiveness and campaign effort have often viewed them as interlinked, particularly in studies of turnout (e.g. Cox and Munger, 1989; Cox et al., 2020). Parties and elites devote more campaign effort and resources in competitive contests: campaigns spend more money, candidates undertake more visits, and parties invest in a greater volume of political advertising when elections are closely fought (Cann and Cole, 2011; Cox and Munger, 1989; Cutts and Johnston, 2015). We recognize this connection, viewing both party campaign efforts and competitiveness as indicators of the intensity of an election campaign. However, whilst they exhibit some conceptual overlap, we assume that campaign effort and competitiveness influence affective polarization via complementary but distinct mechanisms, for several reasons.

First, we view campaign effort as likely influencing affective polarization via the proliferation of party cues and campaign-related information that both raise the salience of party identities and shape voter judgments about parties along partisan lines (Chapman and Palda,

1983; Mackie et al., 2008; Trumm et al., 2017), leading to greater levels of affective polarization. Competitiveness complements this process by triggering greater cognitive engagement with these cues and information (e.g. Kam and Utych, 2011). But competitiveness, we argue, exerts an additional influence on affective polarization by increasing perceived out-group threat (Miller and Conover, 2015), leading both to stronger in-group identification and out-group hostility. This mechanism is distinct from that of campaign effort.

Thus, even holding constant the amount of campaign effort that parties expend in a constituency, there can be variation in competitiveness and hence in how ‘threatening’ out-parties are for voters, which can shape attitudes toward those parties. And holding constant how competitive a constituency-level contest is, we expect variation in how much effort parties choose to expend—and therefore how abundant party cues and party information are. This, again, can shape how parties are perceived.¹

Turning back to our overall research goals, if campaign effort and competitiveness prompt affective polarization over election campaigns, then district-level differences and over-time within-district variation in party activity and the competitiveness of districts provide a potential explanation for within-country variation in (trends in) affective polarization. Exploring the example of Britain—the electoral context under study—reveals this clearly. For example, recent history shows that a voter residing in the constituency of Twickenham will have experienced varying degrees of competitiveness and party campaign effort: in 2015, a usually safe Liberal Democrat seat (a 12,140 vote majority in 2010) was instead won by the Conservative party in a marginal contest (a majority of 2017 votes). Within constituencies, overtime changes in party campaign effort also occurred. For example, in 2015, total candidate campaign spending in Rochester and Strood amounted to £36,578 (adjusted for inflation). In contrast, in 2017, candidates spent a total of £22,911 (adjusted for inflation), representing a decrease of £13,667. These district-level changes from one election to the next potentially drive changes in affective polarization. However, other districts experience little variation in either competitiveness or spending. In East Surrey, for example, the Conservative margin of victory in both 2015 and 2017 was 40.4 %, and the accompanying change in candidate spending was £3233 (from an absolute spend of £22,977 in 2015 to £19,744 in 2017). This between-district heterogeneity in within-district variation in competitiveness and campaign spending provides a potential explanation for the emergence of local differences in affective polarization.

Taken the goals of this study in totality, we seek to assess whether heterogeneity in campaign characteristics—varying campaign effort and electoral competitiveness—prompts variation in the degree to which voters affectively polarize over an election campaign. In doing so, we offer a number of contributions to the extant literature. First, while we know that within-country variation in affective polarization exists (see Bettarelli et al., 2023), the reasons for this variance are unclear. By examining the dynamics of this heterogeneity, we can better understand the causes of affective polarization and the contextual risk factors that increase the likelihood of hostile inter-partisan relations. Second, this study contributes to a burgeoning literature that examines the role of election characteristics in contributing to partisan animus. Previous research in this field has either made the implicit assumption that the polarizing consequences of elections are homogeneous (e.g. Singh and Thornton, 2024) or has examined heterogeneity caused by individual differences (Hernández et al., 2021; Hahm et al., 2023), country-level

factors (Gidron et al., 2023; Hahm et al., 2023) or electoral outcomes (Gidron and Sheffer, 2024; Sheffer, 2020). Consequently, this study represents the first assessment of the role of (heterogeneous) campaign effects as a source of variation in affective polarization over election campaigns. As evidence increasingly demonstrates that elections can polarize voters (Hernández et al., 2021) and recent examples from Brazil and the United States show that conflict on the campaign trail can spill over into political violence (also see Kalmoe and Mason, 2022), understanding the electoral characteristics that prompt polarization is a pressing concern.

3. Data and methods

3.1. Data

To assess the relationship between campaign effects and affective polarization, we examine three national elections in Britain—those held in 2015, 2017 and 2019.² We use data from the British Election Study Internet Panel (henceforth BESIP), a large-*N* panel dataset beginning in 2014 and sampling respondents from the three countries of England, Scotland and Wales.³ Retaining only those individuals who provide non-missing data for our study variables provides us with a dataset of 37,562 person-observations across the three elections. As the data have a panel structure, some survey respondents appear at multiple elections. Supplementary materials Table 1.1 provides information on the distribution of respondents across all three elections and the number of repeated observations in the data.

3.2. Variables

Affective polarization, our outcome variable, is measured using like-dislike scores toward the major British political parties (on a 0–10 scale). For 2015 and 2017, this consists of the Conservatives, Labour, Lib Dems, SNP, Plaid Cymru, UKIP and the Green Party. For 2019, the Brexit Party is included in place of UKIP (reflecting changes in the structure of the British party system during this period). Party like-dislike scores are a widely used, empirically validated approach to measuring affective polarization (e.g. Reiljan, 2020; Wagner, 2021; see also *Gidron et al., 2022*). Importantly, evidence shows that like-dislike scores correlate with other aspects of affective polarization such as social distancing (*Gidron et al., 2022*), meaning that like-dislike scores can be used to make inferences about intergroup relations. Specifically, we utilize the weighted spread score measure proposed by *Wagner (2021)*, with like-dislike scores weighted based on vote share in the corresponding election (using country-specific vote shares from England, Scotland and Wales, rather than UK-wide vote shares). Using this measure provides us with individual-specific affective polarization scores at each time point, which accounts both for the multi-party nature of the British political system and the relative importance of each party at the corresponding election. We detail the process for calculating this measure in Section 1.4 of the supplementary materials, along with aggregate-level means for affective polarization (pre- and post-election day) at each election.

In addition to affective polarization, we analyze its constituent aspects, namely in-party like and out-party dislike. For in-party like, we use the like score corresponding to a respondent’s most-liked party. We use two measures of out-party dislike. First, a single-party measure, measured by the like score corresponding to the primary electoral competitor in the respondent’s constituency. This is the party that finished second if the respondent’s in-party won the local contest, or the party that won

¹ Supporting the proposition that campaign effort and competitiveness are related, but distinct, influences on affective polarization, the correlations between our measures of campaign effort and competitiveness are fairly weak across the three elections under study ($r = 0.136$ for our measures of in-party campaign effort and competitiveness; $r = -0.048$ for out-party campaign effort and competitiveness).

² We focus on these three elections as they are the only three for which data for our key indicators (notably campaign spending) are available at the time of writing.

³ Northern Ireland, the final country within the United Kingdom, is not included in the BESIP.

if the respondent's in-party lost. To assess whether partisan hostility is specific to the primary competitor or generalizes to other out-parties, we also employ a multi-party measure, which is the average like score toward all parties other than the most-liked party. We reverse code these measures of out-party affect so that higher scores correspond to greater dislike toward out-parties.

Our measure of competitiveness is the competitiveness index (CI) score proposed by [Endersby et al. \(2002\)](#). This measure first identifies the effective number of parties in each constituency using the inverse Herfindahl–Hirschman Index, and then uses the *ex post* constituency-level vote shares of these effective parties to produce a constituency-specific measure of competitiveness, adjusted for the effective number of parties in each constituency contest. The index ranges from 0 to 1, with higher scores corresponding to closer elections. We detail the calculation of this measure and its distribution across the sample in supplementary materials Section 1.2. We believe this approach offers a superior measure of competition compared to a margin of victory approach, often employed in studies of competitiveness in single-member plurality systems (see [Blais and Lago, 2009](#)) because it better accounts for the multi-party nature of electoral competition (i.e., the role of third and fourth parties in close races). Furthermore, we believe it provides a more effective means of comparing contests that vary in their effective number of parties (i.e. comparing two-party races with three-party races) than the margin of victory measure. Given the party fragmentation witnessed in British elections over the study period—particularly in 2015 ([Fieldhouse et al., 2020](#))—we believe the competitiveness index offers a superior measure of competitiveness.⁴

A substantial number of constituency races across these three elections were contested meaningfully by three or more parties, and the competitiveness index allows us to account for these multi-party dynamics of political competition. Beyond the margin of victory (i.e. the difference in vote share between the first- and second-placed party), the index accounts for the relative importance of third parties in its calculation. This, we argue, is beneficial for our focus on affective polarization, given that a third party which stands a more meaningful chance of success at the constituency level both provides greater incentives for engagement to its local supporters and poses a greater threat to supporters of other parties, both of which could potentially contribute to affective polarization. We provide a more in-depth discussion of this argument, with examples, in Section 1.2 of the supplementary materials.

To measure campaign effort, we rely on campaign spending as a proxy. Campaign spending is a widely deployed measure of campaign effort that has been shown to exhibit high construct validity ([Fisher et al., 2016](#)) and strongly correlates with other measures of campaign effort ([Fieldhouse and Cutts, 2009](#)). To calculate measures of campaign spending, we use candidate campaign spending data provided by the UK Electoral Commission. Importantly, the Electoral Commission data provide constituency-level spending data by party. By matching this data with information on the political preferences of BESIP respondents, we are able to derive measures of constituency-level in-party and out-party spending for each individual in the data. We define in-party spending as the total amount spent in the respondent's constituency by their most-liked party, using like scores measured at the pre-election wave. If an individual provides an equally high score to multiple parties, we code the in-party as the party that the respondent reports feeling most attached to (measured using a separate survey question). Out-party spending is measured as the total amount spent by all parties other than the in-party in the respondent's constituency (i.e. total constituency spend minus in-party spend).

⁴ However, we do replicate the analysis with the more traditional measure and our findings hold in this alternative specification. See [Section 6.3](#) of this document for an overview and supplementary materials Section 4.4 for a detailed discussion of this analysis.

3.3. Estimation procedure

We use a longitudinal approach to assess the relationship between campaign characteristics and affective polarization. As the BESIP is a panel, it collects data from panel respondents before and after each UK general election. This allows us to measure individual-level affective polarization prior to the beginning of an election campaign and immediately after election day, and thus assess whether campaign effort and competitiveness predict within-person changes in affective polarization over the course of an election period (this change score approach is frequently used in research on the attitudinal consequences of elections, e.g. [Blais et al., 2017](#); [Hooghe and Stiers, 2016](#)). The use of individual-level change in affective polarization from pre- to post-election as the outcome variable means that time-invariant unobservables that influence affective polarization will not bias our estimates for campaign spending and competitiveness ([Allison, 1990](#)). Furthermore, using a change score allows us to rule out the possibility that pre-existing differences in affective polarization between individuals may be biasing the estimates.

For the 2015 election, we use data collected 6 months before the election (at wave 3, September 19th to October 17th 2015) and immediately after the election in May 2015 (wave 6). For 2017, we use data collected 5 months before the election (at wave 10, from November 24th to December 12th 2016) and again immediately after the election in May 2017 (wave 13). Finally, for 2019, we use data collected 6 months before the election (at wave 16, from May 24th to June 18th 2019) and immediately after the election (at wave 19).⁵

We pool data from all three elections (2015, 2017 and 2019) and regress the change in affective polarization from pre- to post-election on competitiveness and campaign spending (in-party and out-party), estimating a single model that includes data from all three general elections. As the BESIP is a panel, the data is clustered with observations nested in individuals, so we fit multilevel linear models with individual-level error terms to account for the non-independence of observations. To account for the influence of (time-invariant) characteristics of each election—such as party-level ideological polarization between elections—all models include election fixed effects. We also include constituency fixed effects. By doing so, we account for all the constituency-specific influences on affective polarization that remain constant over time, which best allows us to isolate the impact of variation in district-level campaign effort and competitiveness. In addition, we include country fixed effects, to account for time-invariant differences in the structure of party competition across countries, which rules out the possibility that differences in party systems across England, Scotland and Wales confound the relationship of interest. We also include a set of socio-demographic controls: age, gender, education, and ethnicity.

Finally, we control for affective polarization scores at the pre-election wave to account for floor and ceiling effects. When comparing changes in an outcome variable between individuals, convergence to the mean due to ceiling effects is a potential issue ([Singh et al., 2012](#)). In other words, highly polarized individuals are less likely to report large increases in affective polarization due to their pre-election position on the outcome scale. Given that we expect consistency in competitiveness and party campaign effort within certain constituencies across elections, ceiling effects (unless accounted for) could obscure changes in affective

⁵ The BESIP also contains pre-election waves that are temporally closer to election day (wave 4 for 2015; wave 11 for 2017; and wave 17 for 2019). However, as these three waves all begin following the initial announcement of the election, they are likely subject to post-treatment effects, given that the announcement of the election provides an initial cue or trigger that amplifies the salience of partisan identity and thus drives affective polarization. Consequently, we employ earlier waves that avoid potential post-treatment bias in the estimation. As a robustness test, we replicate the analysis with a more proximate pre-election wave. Results are presented in Section 4.5 of the supplementary materials.

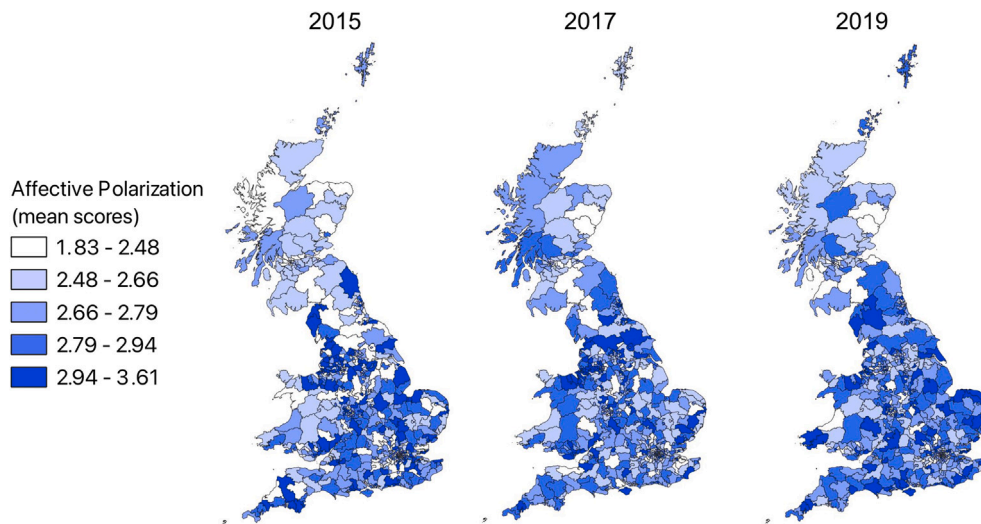


Fig. 1. Constituency-level mean affective polarization: BESIP 2015–2019. Affective polarization scores are calculated using the weighted spread score approach described in Section 3. At each election, constituencies with less than 20 person-observations were excluded.

polarization in the constituencies that are most marginal and receive the most party effort. We thus control for pre-election affective polarization to rule out this concern.

This results in the following model:

$$\Delta AP_{it} = CI_{jt} + Spend_{jt} + Election_t + Constituency_i + x_{it} + AP_{it-1} + \varepsilon_i \quad (1)$$

These models provide an estimate of 1) whether, relative to those in less competitive districts, individuals in competitive contests become more affectively polarized over the course of an election campaign and 2) whether, relative to those in districts that receive less campaign effort from parties, individuals residing in high-effort districts become more affectively polarized over an election campaign. All models are estimated using the *lme4* package in R version 4.5.1 (Bates et al., 2015; R Core Team, 2021).

4. The British case

Before presenting the results of our analyses, we provide greater detail on the electoral context under study. We believe that the case of Britain from 2015 to 2019 offers an excellent test-bed for our hypotheses for a number of reasons. First, the presence of significant between-constituency variation in both competitiveness and spending. As with other SMDP systems, British elections are characterized by the combination of highly competitive marginal seats alongside ‘safe’ seats in which one party dominates.⁶ This between-constituency variation also extends to campaign efforts. British political parties employ campaign effort selectively, predominantly targeting competitive but winnable seats, with safe seats (either held by a member of their own party or an opposition party) less likely to be the subject of campaign efforts (Fisher et al., 2016). For example, the Labour Party spent £13,307 in the Calder Valley constituency during the 2019 election campaign, nearly double their constituency-level average spend of £7423. In contrast, in Liverpool Walton, a consistently safe Labour constituency, the party spent only £4846 on the campaign. Thus, Labour supporters residing in Liverpool Walton will have been exposed to less in-party campaign effort in 2019 than Labour voters in Calder Valley.

⁶ Descriptive analysis of the competitiveness index (coded 0–1) illustrates this. For 2015, the index has a standard deviation of 0.15 and minimum and maximum constituency scores of 0.02 and 0.94. In 2017, the index *SD* is 0.14, with minimum and maximum values of 0.22 and 0.95. In 2019, the index *SD* is 0.17, with minimum and maximum scores of 0.22 and 0.92.

Second, the occurrence of three national elections within a 6-year time span provides us with a unique research context that encompasses a broad range of national election conditions.⁷ The 2015, 2017 and 2019 elections varied slightly in their outcomes (majority Conservative governments in 2015 and 2019; a minority Conservative government in 2017) but moreso in the nature of constituency-level contests in each election. Comparing the effective number of parties in each constituency across each election, which represents the number of parties in each constituency that won enough votes to be a substantively important competitor, reveals variation across elections (see Section 1.2 of the supplementary materials). 2015 was mainly constituted by three-party contests (77.8 % of contests contained three effective parties), 2017 was made up predominantly of two-party contests (76.7 % of contests contained two effective parties) and 2019 was something of a middle ground (57 % of races contested by two effective parties, 40.2 % by three). These varying conditions across elections provide a range of electoral conditions to test our hypotheses across.

Third, an assumption of our research design is that local campaign effects will matter for affective polarization, beyond the national-level campaign. Evidence indicates that local campaigns are an important influence on voter behavior at national elections in Britain (Whiteley and Seyd, 1994; Whiteley and Seyd, 2003), supporting this assumption. In addition, analysis of local spending at the national elections under study indicates that the majority of local campaign effort is devoted to advertising and leafleting (Electoral Commission, 2016; Power, 2025), consistent with our characterization of campaign effort as a combination of party cues and information provision.

Finally, Britain exhibited much constituency-level variation in affective polarization over the study period. Using British Election Study Internet Panel data, Fig. 1 above depicts constituency-level mean affective polarization scores in the 2015, 2017 and 2019 post-election waves. These comparisons reveal variation both between constituencies, and within the same constituencies over time—two important assumptions for our estimation strategy.

In sum, we view our case as a uniquely advantageous context in which to test our hypotheses, given that the occurrence of three national elections within a 6-year time span provides sufficient within-

⁷ It is important to highlight that constituency boundaries did not change within this period. The constituency boundaries used in these elections were implemented in 2010 and remained in place until 2024.

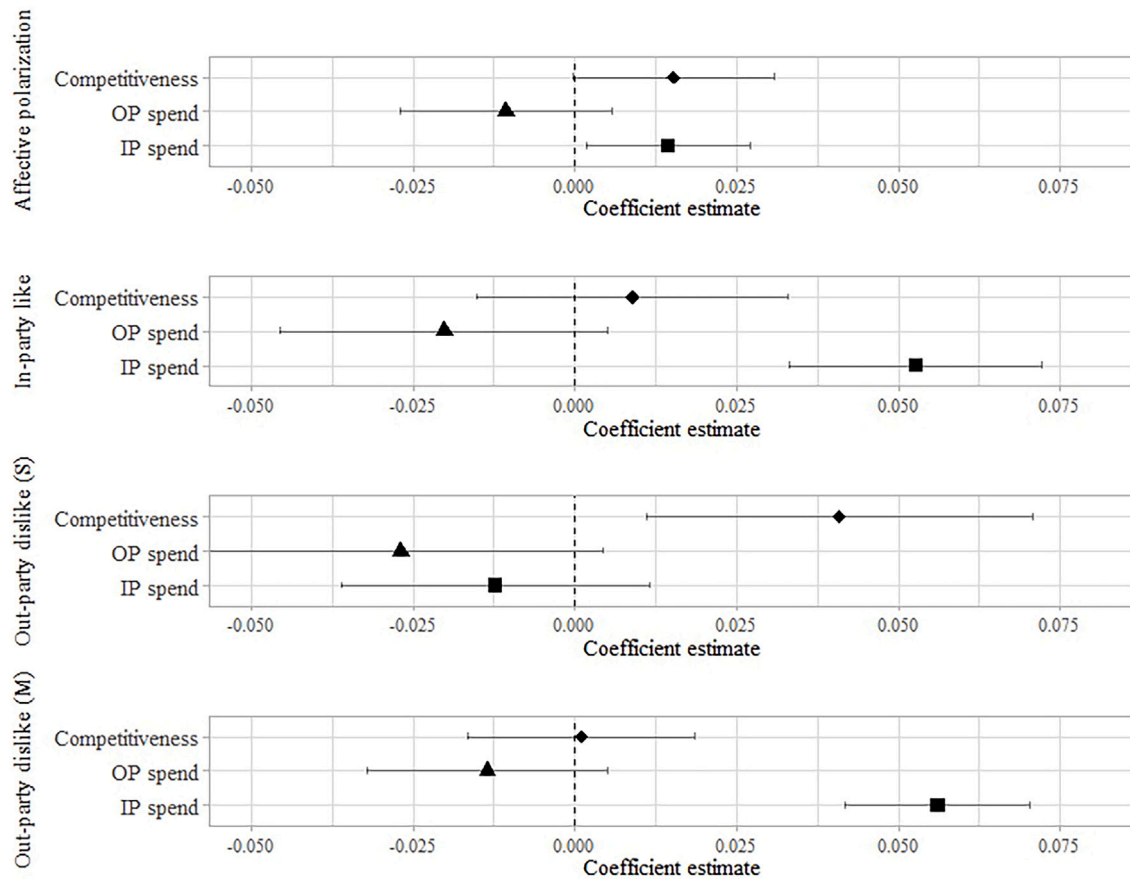


Fig. 2. The regression of affective polarization on district-level campaign effort and competitiveness: 2015–2019. *Note:* Points are standardized linear fixed effects coefficients with 95 % confidence intervals. Out-party dislike (S) refers to the model in which the closest-competitor measure of out-party dislike is the dependent variable; Out-party dislike (M) refers to the model in which the multi-party measure of out-party dislike is the dependent variable. Affective polarization model $N = 37,562$; in-party like model $N = 37,562$; out-party like single model $N = 34,824$; out-party like multi model $N = 37,562$.

and between-constituency variation in campaign characteristics to test the consequences of local campaign effects for affective polarization. Regarding external validity, the SMDP system used for British elections likely makes any findings more generalizable to other national contexts that use similar SMDP systems (such as Australia, the USA or Canada) than proportional electoral systems. We return to this issue in the discussion.

5. Results

To begin, we regress changes in affective polarization on competitiveness and campaign spending, including election and constituency fixed effects and controls. The competitiveness and campaign spending measures are standardized before inclusion in the model, so the coefficients represent the estimated effect of a one-standard deviation increase in the corresponding measure. The results are represented in Fig. 2 (full model output is available in the supplementary materials, Table 2). Results indicate that in-party spending is positively associated with affective polarization ($b = 0.014$, $SE = 0.006$, $p < 0.05$). Voters who reside in constituencies in which their in-party spends more campaign money grow more affectively polarized over the course of the election campaign, confirming *H1*. Out-party spending in contrast, is unrelated to affective polarization ($b = -0.011$, $SE = 0.008$, $p = 0.193$). Turning to the consequences of party competition, results indicate that competitiveness is positively associated with affective polarization, albeit at $p < 0.1$ ($b = 0.015$, $SE = 0.008$, $p < 0.1$).

These results provide evidence that in-party campaign spending contributes to affective polarization during election periods, and borderline evidence that competitiveness exerts a similar influence. Next, recognising the possibility that in-party spending, out-party spending and competitiveness could have heterogeneous consequences for in-party like and out-party dislike, we assess the relationship between local campaign effects and these different dimensions of affective polarization.

First, we assess the relationship between campaign spending, competitiveness and in-party like. The results, which are summarized in the second panel of Fig. 1, indicate that in-party spending is positively and significantly associated with in-party like ($b = 0.053$, $SE = 0.01$, $p < 0.001$). The more an individual's in-party spends in their local constituency during the election campaign, the more that individual grows to like their in-party over the course of the campaign. Neither competitiveness nor out-party spending is associated with in-party like. Full model results are presented in Table 2, column II, of the supplementary materials.

Second, we examine the relationship between spending, competitiveness and out-party dislike, which is displayed in the third and fourth panels of Fig. 1. First focusing on the change in dislike toward the largest competitor in the respondent's district (the party that won the district if the respondent's party lost, and the party that came second if the respondent's party won), results indicate that competitiveness is positively associated with changes in dislike toward this party ($b = 0.041$, $SE = 0.015$, $p < 0.01$). In-party spending is unrelated to dislike toward the closest competitor, and out-party spending is negatively associated with

this outcome, albeit at $p < 0.1$ ($b = 0.027$, $SE = 0.016$). Complete results from this model are provided in supplementary materials Table 2, column III.

Next, we assess whether local campaign effects are focused on the closest competitor or whether they generalize to all out-parties, using a measure of multi-party dislike (average dislike toward all parties, other than the in-party). The results of this model indicate that in-party spending is positively associated with general out-party dislike ($b = 0.056$, $SE = 0.007$, $p < 0.001$). Out-party spending and competitiveness are unrelated to this outcome. Full model results are presented in Table 2 of the supplementary materials, in column IV.

In sum, results indicate that when a party expends greater campaign funds (and hence effort) in a constituency, supporters of this party become more affectively polarized. And the consequences of campaign effort are two-directional, with supporters displaying both greater liking toward their party and greater dislike toward other parties. Regarding the size of this effect, a one standard deviation increase in in-party spending is associated with: an increase of 0.015 standard deviations in affective polarization; an increase of 0.031 standard deviations in in-party liking; and an increase of 0.048 standard deviations in out-party dislike. This level of change is similar to that observed in prior research on electorally induced affective polarization (e.g. Bassan-Nygate and Weiss, 2022; Singh and Thornton, 2024). Furthermore, because our estimation strategy compares within-person change between individuals, these estimates represent increases in affective polarization on top of the general increases in affective polarization that we would expect the national-level campaign to induce across the whole electorate (e.g. Hernández et al., 2021).

Second, results indicate that competitiveness is positively associated with affective polarization, and that this is driven by changes in dislike toward the closest electoral competitor. So, all else being equal, an individual who resides in a more competitive constituency grows to dislike the party that they predominantly compete with more than individuals who reside in less competitive constituencies. A one standard deviation increase in the competitiveness index is associated with an increase of 0.023 standard deviations in dislike toward the closest competitor. Again, this effect is of a similar magnitude to prior estimates of the consequences of elections for affective polarization (Bassan-Nygate and Weiss, 2022; Singh and Thornton, 2024).

6. Further analysis

6.1. Testing the mechanisms

Not all voters possess long-term party attachments. Whilst some are committed partisans, others engage with party politics only during election times (Singh and Thornton, 2019). The processes by which we argue in-party campaign spending affectively polarizes voters over the course of an election—party cues and the directionally motivated acceptance of in-party campaign information—are identity-based processes (e.g. Mullen et al., 1992; Slothuus and de Vreese, 2010). Given this, we would expect the relationship between spending and affective polarization to be larger among voters who possess a pre-existing party identity. Furthermore, in competitive contests, heightened voter engagement and perceptions of out-group threat are potentially more likely among those with pre-existing partisanship, for whom in-group success will be more important. While we find only marginal evidence for a main effect of competitiveness on affective polarization, this relationship may be stronger among partisans.

In light of these expectations, we test whether the relationships between (a) campaign effort and affective polarization and (b) competitiveness and affective polarization are larger among individuals with a pre-existing party identity (measured during the pre-election period) than among non-partisans (with partisanship measured during the pre-election period). These tests provide a means of understanding the mechanisms that drive the relationships of interest.

Fig. 3 plots the average marginal effects of in-party campaign effort, out-party campaign effort and competitiveness on affective polarization among partisans and non-partisans. Complete results and a summary of our modelling strategy are presented in supplementary materials Section 3.1.⁸

Regarding in-party spending, results provide support for our expectations. The interaction term is positive and significant at $p < 0.01$ ($b = 0.046$, $SE = 0.016$), indicating that partisanship moderates the relationship between in-party spending and affective polarization. Among partisans, we find that the relationship between in-party campaign spending and affective polarization is positive ($AME = 0.026$, $SE = 0.006$, $p < 0.001$) while among non-partisans, in-party spending is negatively associated with affective polarization, albeit not statistically significant. Further analysis (presented in Section 3.1 of the supplementary materials) reveals that in-party spending leads to both greater levels of in-party liking and out-party disliking among partisans, but not among non-partisans. Consistent with the logic of motivated reasoning (Slothuus and de Vreese, 2010), it appears that attitudes toward parties are influenced by in-party campaign efforts only when individuals possess a pre-existing attachment to the in-party.

In contrast, we find that competitiveness is positively associated with affective polarization among both partisans ($AME = 0.016$, $SE = 0.008$, $p < 0.05$) and non-partisans ($AME = 0.025$, $SE = 0.015$, $p = 0.11$). That the estimate for non-partisans is less precise is unsurprising given that this group is much smaller (only around 15 % of individuals do not express a party attachment). This suggests that the mechanisms by which competitiveness leads to affective polarization are not specific to partisans. Furthermore, analysis in supplementary materials Section 3.1 shows that competitiveness leads to greater levels of dislike toward the closest competitor among both those with a pre-existing partisan identity ($b = 0.044$, $SE = 0.015$, $p < 0.01$) and those without ($b = 0.065$, $SE = 0.031$, $p < 0.05$). This provides evidence that the process by which competitiveness leads to out-party dislike via perceived out-group threat is one that generalizes across those who participate in an election, not just those with a pre-existing party identity.

6.2. The interaction of competitiveness and campaign spending

Assessing standalone relationships reveals that in-party campaign spending is positively associated with affective polarization. In a follow-up test, we assess whether competitiveness moderates the relationship between in-party campaign effort and affective polarization. This model, which we present and discuss in detail in Section 3.2 of the supplementary materials, provides evidence that competitiveness moderates the relationship between in-party spending and affective polarization. Specifically, results suggest that in highly competitive electoral contests, the relationship between in-party campaign spending and affective polarization is attenuated. We suggest that greater attention and engagement from voters in highly competitive contests (Kam and Utych, 2011; Roy et al., 2015) provides a potential ceiling effect on the impact of in-party campaign effort. If voters are already seeking out information about the campaign, then they will already be receiving the party cues and information that party campaign effort would otherwise provide, attenuating the influence of in-party spending on affective polarization.

6.3. Robustness tests

In Section 4 of the supplementary materials, we undertake a series of replications with alternative model specifications in order to test the robustness of our conclusions. First, we address the question of whether potential multicollinearity between our measures of campaign spending

⁸ Given the focus on partisanship in these models, we use respondent partisanship (measured pre-election) to define the in-party for partisans in these models. Further discussion of changes to our estimation strategy is provided in supplementary materials Section 3.1.

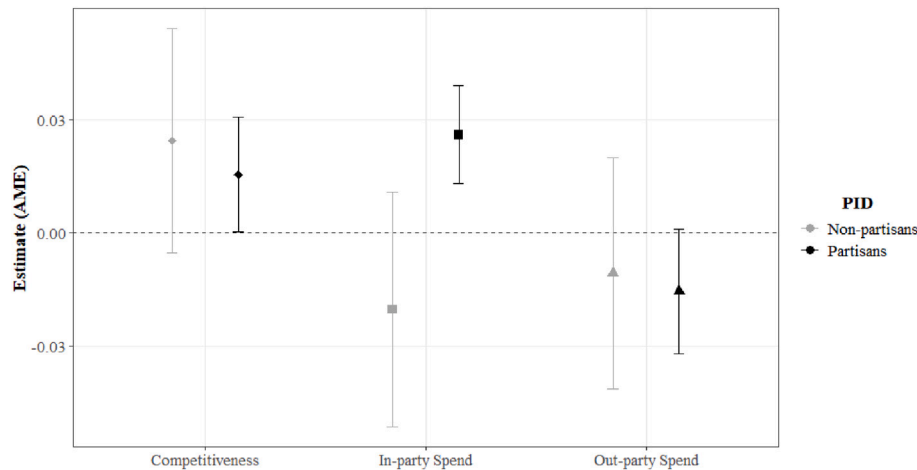


Fig. 3. Average marginal effects of campaign effort and competitiveness on affective polarization by partisanship. *Note:* Points are average marginal effects with 95 % confidence intervals, representing the estimated change in affective polarization associated with one standard deviation increase in the predictor. Model $N = 39,448$.

and competitiveness could be biasing our results. To do so, we fit separate models for campaign spending and competitiveness, but find no evidence that the results change when the relationship between affective polarization and each of our key constructs is measured separately. Results are presented in supplementary materials Section 4.1.

Second, we replicate the analysis with a sub-sample of constituencies (and individuals), excluding those constituencies in which a far-right candidate was a meaningful competitor. Given evidence that affective polarization between partisans is especially hostile when it involves far-right parties (Hartevelde et al., 2022), it is important to determine that changes in far-right presence or support are not confounding the observed relationships between campaign effort, competitiveness and affective polarization. Results presented in supplementary materials Section 4.2 are broadly similar when far-right parties are excluded, providing evidence that factors associated with far-right electoral contestation are not confounding the relationships of interest.

Third, given that observations are clustered both within individuals and within constituencies, we assess whether constituency-level clustering biases our results by fitting multilevel models with a two-level error structure to account for clustering at both the individual and constituency levels. Results remain unchanged with this alternative model specification, and can be found in supplementary materials Section 4.3.

Fourth, given the debate over the best approach to measure district-level competitiveness (e.g. Cox et al., 2020), we replicate the analysis with an alternative measure of competitiveness (the distance between the first and second-placed party in each constituency, measured in raw votes). Fifth, we replicate the analysis with self-reported measures of in-party and out-party contact, as an alternative to actual party spending. Results, which are presented in supplementary materials Sections 4.4 and 4.5, match those of the main models, suggesting that our findings are not an artefact of our approach to measuring competitiveness or campaign effort.

Sixth, in supplementary materials Section 4.6, we fit models which use a pre-election baseline wave that is more proximate to the election than the baseline wave used in the main models. Whilst we believe this approach offers an inferior strategy to the one used in the main models due to concerns about post-treatment bias, results indicate that in-party spending remains positively associated with affective polarization, although the size of this association is reduced relative to the main models. Competitiveness is positively associated with affective polarization in these models, driven largely by dislike toward the closest competitor. The reduction in the magnitude of the coefficient for in-party campaign spending is potentially consistent with the idea that the more proximate pre-election wave is post-treatment to the campaign.

Finally, there is an ongoing debate in the literature over the relative merits of different measures of affective polarization (e.g. Reiljan, 2020; Wagner, 2021). Given this, and to ensure that our results are not an artefact of the specific measure of affective polarization that we have chosen, we replicate the analysis with the affective polarization index proposed by Reiljan (2020) in supplementary materials Section 4.7. Results are robust to this alternative measure of affective polarization.

7. Discussion

Our goal with this paper was to examine the effect of local variation in campaign effects—specifically electoral competitiveness and party campaign effort—on affective polarization. A number of studies have previously shown that elections make partisan identity more salient and in this way foster affective polarization (Hernández et al., 2021; Singh and Thornton, 2024). However, we argue these studies overlook important heterogeneity in the nature of elections, because there is much variation in the experience of campaigns within countries, especially in SMDP systems where parties allocate scarce resources to some districts and invest less in others and there are important within-country differences in how closely fought elections are.

Analyzing election survey data that was collected around the 2015, 2017 and 2019 British elections, our results can be summarized into three central findings. First, we find evidence that in-party campaign effort is positively associated with changes in affective polarization over an election campaign. When a party exerts more effort (via campaign spending) in a local campaign, its supporters grow more affectively polarized than in constituencies which receive less campaign effort. Further analysis reveals that in-party spending leads both to greater levels of in-party like and out-party dislike. As we would expect, as a party spends more money and provides a greater quantity of in-group cues and partisan campaign information to its supporters, those supporters grow to like the party more. And our results suggest that this effort has the additional consequence of amplifying dislike toward out-parties. These findings are consistent with the idea that in-party campaign effort both primes party identity via in-group identity cues and influences perceptions of the in-party and out-parties via (potentially partial) political information, leading to greater levels of affective polarization over the campaign. Follow-up analysis, which indicates that the relationship between in-party spending and affective polarization is limited to partisans (i.e. individuals with a partisan identity prior to the onset of the election campaign), supports these proposed mechanisms.

Second, we find evidence that constituency-level competitiveness is associated only with increases in dislike toward the closest competitor

party in one's constituency. In other words, the closer the contest, the more a voter grows to dislike the party that offers the largest competition in their local race. Intuitively, it appears that voters target hostility toward the party that represents the strongest threat to their desired electoral outcome, consistent with the idea that closer races lead to greater negativity toward out-parties (Miller and Conover, 2015). Follow-up analysis indicates that competitive contests lead to greater levels of dislike toward the closest competitor among all voters, suggesting that all voters perceive the out-party as threatening in competitive contests, not just committed partisans. Elections are focusing events that prompt previously disengaged individuals to pay attention to, and involve themselves in, party politics (Singh and Thornton, 2019). It is perhaps this quality of elections which means that the dynamics of a closely fought contest shape out-group attitudes among both partisans and non-partisans alike.

Third, findings indicate that out-party spending is unrelated to affective polarization and its constituent dimensions. This provides support for the idea that in-party loyalty acts as a filter through which voters process party campaign information and cues. Put differently, the null effects for out-party spending suggest that when campaign efforts originate from opposing parties, voters ignore or dismiss those campaign efforts, consistent with the idea that voters are overwhelmingly influenced by party cues and frames when they originate from their favoured party (e.g. Slothuus and de Vreese, 2010; Bolsen et al., 2014) or that voters filter their information diets in order to selectively expose themselves to attitudinally congruent information alone (Iyengar and Hahn, 2009; Peterson et al., 2021).

These findings offer a number of contributions to the extant literature on affective polarization. First, whilst prior research has established the existence of within-country differences in affective polarization (Bettarelli et al., 2023), investigation of possible factors that explain this local variation has to the best of our knowledge yet to occur. This article addresses this gap by providing evidence for local variation in campaign effects as a source of within-country variation in affective polarization. Second, previous research on election campaigns and affective polarization has focused on individual differences (Hernández et al., 2021; Hahm et al., 2023), country-level factors (Gidron et al., 2023; Hahm et al., 2023) or electoral outcomes (Gidron and Sheffer, 2024; Sheffer, 2020) as sources of individual-level variation in affective polarization during election campaigns. In demonstrating the importance of local campaign effects as another source of variation in the polarizing consequences of election campaigns, we provide evidence of an additional mechanism through which elections can polarize voters. Finally, this article offers a substantive contribution. Given evidence that elections polarize voters (Hernández et al., 2021), and recent history from the US and Brazil showing that elections can act as inflection points in the polarization of electorates, understanding how and why elections polarize grows ever more important. Our findings contribute toward that goal.

Our work also suffers from a number of limitations. First, while our work sheds light on the consequences of district-level competitiveness for affective polarization, the local campaign is only part of the electoral context to which citizens are exposed during an election campaign. To the extent that politics and campaigns are nationalized (Caramani, 2004), citizens might be much more influenced by dynamics and by how competitive the election is at the national level. To be sure, our analyses account for such nation-wide effects through the inclusion of election fixed effects, but the design does not allow us to estimate the relative importance of district versus nation-wide campaign effects. Second, our analysis is restricted to a single context: Britain. This inevitably limits the potential for generalization. The SMDP electoral system that is used in Britain is also particular and might be inherently more polarizing than proportional electoral rules because only the first party in a locality wins a seat. Relatedly, some of the dynamics that we observe—such as the finding that more marginal contests produce greater dislike toward the closest electoral competitor in one's constituency—might not hold in multi-member proportional systems in which multiple parties

can win seats. However, this characteristic of the British system is also a strength, resulting in a large N of districts and allowing us to study the polarizing effects of elections through a geographical lens. Further research, extending these tests to proportional systems, is necessary to clarify the generalizability of the findings.

Despite these limitations, our work provides important insights into the sources of affective polarization. In line with work that has leveraged the temporal proximity to elections (e.g. Hernández et al., 2021), our results indicate that when partisan identities are more salient, affective polarization increases. We contribute to this literature by demonstrating that not only the event of an election, but also the characteristics of that election drive affective polarization. Furthermore, in showing that local-level competitiveness correlates with the strength of (an increase in) affective polarization, our work points to a potential source of important within-country differences in affective polarization (Bettarelli et al., 2023). Elections are lived and experienced very differently across a country, because local electoral realities differ strongly. This variation, we show, shapes how affectively polarized citizens become.

CRedit authorship contribution statement

Justin Robinson: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Ruth Dassonneville:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Justin Robinson reports that financial support was provided by the UK Economic and Social Research Council.

Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.electstud.2025.103042.

Data availability

Replication data is available at: <https://doi.org/10.7910/DVN/AY8RT8>.

References

- Allison, P.D., 1990. Change scores as dependent variables in regression analysis. *Soc. Methodol.* 20, 93–114. <https://doi.org/10.2307/271083>
- Bassan-Nygate, L., Weiss, C.M., 2022. Party competition and cooperation shape affective polarization: evidence from natural and survey experiments in Israel. *Comp. Polit. Stud.* 55 (2), 287–318.
- Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67, 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bettarelli, L., Reiljan, A., Van Haute, E., 2023. A regional perspective to the study of affective polarization. *Eur. J. Polit. Res.* 62 (2), 645–659.
- Bettarelli, L., Van Haute, E., 2022. Regional inequalities as drivers of affective polarization. *Reg. Stud. Reg. Sci.* 9 (1), 549–570.
- Bisgaard, M., Slothuus, R., 2018. Partisan elites as culprits? How party cues shape partisan perceptual gaps. *Am. J. Polit. Sci.* 62 (2), 456–469. <https://doi.org/10.1111/ajps.12349>
- Blais, A., Morin-Chassé, A., Singh, S.P., 2017. Election outcomes, legislative representation, and satisfaction with democracy. *Party Polit.* 23 (2), 85–95. <https://doi.org/10.1177/1354068815583200>
- Blais, A., Lago, L., 2009. A general measure of district competitiveness. *Electoral Stud.* 28 (1), 94–100.
- Bolsen, T., Druckman, J.N., Cook, F.L., 2014. The influence of partisan motivated reasoning on public opinion. *Polit. Behav.* 36, 235–262. <https://doi.org/10.1007/s11109-013-9238-0>
- Boxell, L., Gentzkow, M., Shapiro, J.M., 2024. Cross-country trends in affective polarization. *Rev. Econ. Stat.* 106 (2), 557–565.
- Cann, D.M., Cole, J.B., 2011. Strategic campaigning, closeness, and voter mobilization in U.S. Presidential elections. *Electoral Stud.* 30 (2), 344–352.
- Caramani, D., 2004. *The Nationalization of Politics: The Formation of National Electorates and Party Systems in Western Europe*. Cambridge University Press.
- Chapman, R.G., Palda, K.S., 1983. Electoral turnout in rational voting and consumption perspectives. *J. Consum. Res.* 9 (4), 337–346.
- Cox, G.W., Fiva, J.H., Smith, D.M., 2020. Measuring the competitiveness of elections. *Polit. Anal.* 28 (2), 168–185.

- Cox, G.W., Munger, M.C., 1989. Closeness, expenditures, and turnout in the 1982 U.S. House elections. *Am. Polit. Sci. Rev.* 83 (1), 217–231.
- Cutts, D., Johnston, R., 2015. Constituency campaigning intensity and its impact at first and second-order elections: Wales, 2010 and 2011. *Electoral Stud.* 39, 85–101.
- Downs, A., 1957. *An Economic Theory of Democracy*. Harper and Row, New York.
- Druckman, J.N., Petersen, E., Slothuus, R., 2013. How elite partisan polarization affects public opinion formation. *Am. Polit. Sci. Rev.* 107 (1), 57–79.
- Duffy, J., Tavits, M., 2008. Beliefs and voting decisions: a test of the pivotal voter model. *Am. J. Polit. Sci.* 52 (3), 603–618.
- Electoral Commission, 2016. UK Parliamentary General Election 2015: Campaign Spending Report. Technical report.
- Endersby, J.W., Galatas, S.E., Rackaway, C.B., 2002. Closeness counts in Canada: voter participation in the 1993 and 1997 federal elections. *J. Polit.* 64 (2), 610–631.
- Falkenberg, M., Galeazzi, A., Torricelli, M., Di Marco, N., Larosa, F., Sas, M., Mekacher, A., Pearce, W., Zollo, F., Quattrocchi, W., Baronchelli, A., 2022. Growing polarization around climate change on social media. *Nat. Clim. Change* 12, 1–8.
- Fasching, N., Iyengar, S., Lelkes, Y., Westwood, S.J., 2024. Persistent polarization: the unexpected durability of political animosity around US elections. *Sci. Adv.* 10 (36).
- Fieldhouse, E., Cutts, D., 2009. The effectiveness of local party campaigns in 2005: combining evidence from campaign spending and agent survey data. *Br. J. Polit. Sci.* 39 (2), 367–388. <https://doi.org/10.1017/S0007123409000726>
- Fieldhouse, E., Green, J., Evans, G., Mellon, J., Prosser, C., Schmitt, H., Eijk, C.V.D., 2020. *Electoral Shocks: The Volatile Voter in a Turbulent World*. Oxford University Press.
- Fieldhouse, E., Green, J., Evans, G., Mellon, J., Prosser, C., Bailey, J., de Geus, R., Schmitt, H., van der Eijk, C., Griffiths, J., Perrett, S., 2024. British election study internet panel waves 1–29.
- Fisher, J., Fieldhouse, E., Johnston, R., Pattie, C., Cutts, D., 2016. Is all campaigning equally positive? The impact of district level campaigning on voter turnout at the 2010 British General Election. *Party Polit.* 22 (2), 215–226.
- Garcia, S.M., Tor, A., Schiff, T.M., 2013. The psychology of competition: a social comparison perspective. *Perspect. Psychol. Sci.* 8 (6), 634–650. <https://doi.org/10.1177/1745691613504114>
- Garzia, D., da Silva, F.F., Maye, S., 2023. Affective polarization in comparative and longitudinal perspective. *Public Opin. Q.* 87 (1), 219–231.
- Gidengil, E., Stolle, D., Bergeron-Boutin, O., 2022. The partisan nature of support for democratic backsliding: a comparative perspective. *Eur. J. Polit. Res.* 61 (4), 901–929.
- Gidron, N., Adams, J., Horne, W., 2023. Who dislikes whom? Affective polarization between pairs of parties in western democracies. *Br. J. Polit. Sci.* 53 (3), 997–1015.
- Gidron, N., Sheffer, L., 2024. Differentiating the sources of post-election partisan affect warming. *Eur. J. Polit. Res.* 63 (3), 1155–1174.
- Gidron, N., Sheffer, L., Mor, G., 2022. Validating the feeling thermometer as a measure of partisan affect in multi-party systems. *Electoral Stud.* 80.
- Graham, M.H., Svobik, M.W., 2020. Democracy in America? Partisanship, polarization, and the robustness of support for democracy in the United States. *Am. Polit. Sci. Rev.* 114 (2), 392–409.
- Hahn, H., Hilpert, D., König, T., 2023. Divided by Europe: affective polarisation in the context of European elections. *West Eur. Polit.* 46 (4), 705–731.
- Harteveld, E., 2021. Ticking all the boxes? A comparative study of social sorting and affective polarization. *Electoral Stud.* 72.
- Harteveld, E., Mendoza, P., Rooduijn, M., 2022. Affective polarization and the populist radical right: creating the hate? *Gov. Oppos.* 57 (4), 703–727.
- Hernández, E., Anduiza, E., Rico, G., 2021. Affective polarization and the salience of elections. *Electoral Stud.* 69.
- Hobolt, S.B., Leeper, T.J., Tilley, J., 2021. Divided by the vote: affective polarization in the wake of the Brexit referendum. *Br. J. Polit. Sci.* 51 (4), 1476–1493.
- Hooghe, M., Stiers, D., 2016. Elections as a democratic linkage mechanism: how elections boost political trust in a proportional system. *Electoral Stud.* 44, 46–55. <https://doi.org/10.1016/j.electstud.2016.08.002>
- Huddy, L., Mason, L., Aarøe, L., 2015. Expressive partisanship: campaign involvement, political emotion, and partisan identity. *Am. Polit. Sci. Rev.* 109, 1–17.
- Iyengar, S., Sood, G., Lelkes, Y., 2012. Affect, not ideology: a social identity perspective on polarization. *Public Opin. Q.* 76 (3), 405–431.
- Iyengar, S., Hahn, K.S., 2009. RED media, blue media: evidence of ideological selectivity in media use. *J. Commun.* 59 (1), 19–39. <https://doi.org/10.1111/j.1460-2466.2008.01402.x>
- Iyengar, S., Westwood, S.J., 2015. Fear and loathing across party lines: new evidence on group polarization. *Am. J. Polit. Sci.* 59 (3), 690–707.
- Kalmoe, N.P., Mason, L., 2022. *Radical American Partisanship: Mapping Violent Hostility, Its Causes, and the Consequences for Democracy*. The University of Chicago Press, Chicago.
- Kam, C.D., 2006. Political campaigns and open-minded thinking. *J. Polit.* 68 (4), 931–945. <https://doi.org/10.1111/j.1468-2508.2006.00481.x>
- Kam, C.D., Utych, S.M., 2011. Close elections and cognitive engagement. *J. Polit.* 73 (4), 1251–1266.
- Karp, J.A., Banducci, S.A., Bowler, S., 2008. Getting out the vote: party mobilization in a comparative perspective. *Br. J. Polit. Sci.* 38 (1), 91–112.
- Kunda, Z., 1990. The case for motivated reasoning. *Psychol. Bull.* 108 (3), 480–498. <https://doi.org/10.1037/0033-2909.108.3.480>
- Levine, D.K., Palfrey, T.R., 2007. The paradox of voter participation? A laboratory study. *Am. Polit. Sci. Rev.* 101, 143–158.
- Mackie, D.M., Smith, E.R., Ray, D.G., 2008. Intergroup emotions and intergroup relations. *Soc. Pers. Psychol. Compass* 2 (5), 1866–1880.
- Malhotra, D., 2010. The desire to win: the effects of competitive arousal on motivation and behavior. *Organ. Behav. Hum. Decis. Process.* 111 (2), 139–146. <https://doi.org/10.1016/j.obhdp.2009.11.005>
- Matthes, J., 2012. Exposure to counterattitudinal news coverage and the timing of voting decisions. *Commun. Res.* 39 (2), 147–169. <https://doi.org/10.1177/0093650211402322>
- Miller, P.R., Conover, P.J., 2015. Red and blue states of mind: partisan hostility and voting in the United States. *Polit. Res. Q.* 68 (2), 225–239. <https://doi.org/10.1177/1065912915577208>
- Mullen, B., Brown, R., Smith, C., 1992. Ingroup bias as a function of salience, relevance, and status: an integration. *Eur. J. Soc. Psychol.* 22 (2), 103–122.
- Nir, L., Druckman, J.N., 2008. Campaign mixed-message flows and timing of vote decision. *Int. J. Public Opin. Res.* 20 (3), 326–346. <https://doi.org/10.1093/ijpor/edn035>
- Oakes, P.J., 1987. The salience of social categories. In: *Rediscovering the Social Group: A Self-Categorization Theory*. Blackwell.
- Petersen, M.B., Skov, M., Serritzlew, S., Ramsøy, T., 2013. Motivated reasoning and political parties: evidence for increased processing in the face of party cues. *Polit. Behav.* 35 (4), 831–854. <https://doi.org/10.1007/s11109-012-9213-1>
- Peterson, E., Goel, S., Iyengar, S., 2021. Partisan selective exposure in online news consumption: evidence from the 2016 presidential campaign. *Polit. Sci. Res. Methods* 9 (2), 242–258.
- Power, S., 2025. Candidate Spending in the 2024 General Election. Technical report UK Parliament.
- R Core Team, 2021. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Reiljan, A., 2020. Fear and loathing across party lines' (also) in Europe: affective polarisation in European party systems. *Eur. J. Polit. Res.* 59 (2), 376–396.
- Reiljan, A., Garzia, D., Silva, F.F.D., Trechsel, A.H., 2024. Patterns of affective polarization toward parties and leaders across the democratic world. *Am. Polit. Sci. Rev.* 118 (2), 654–670.
- Riker, W.H., Ordeshook, P.C., 1968. A theory of the calculus of voting. *Am. Polit. Sci. Rev.* 62, 25–42.
- Rodriguez, C.G., Gadarian, S.K., Goodman, S.W., Pepinsky, T.B., 2022. Morbid polarization: exposure to COVID-19 and partisan disagreement about pandemic response. *Polit. Psychol.* 43 (6), 1169–1189.
- Roy, J., Singh, S.P., Fournier, P., Andrew, B., 2015. An experimental analysis of the impact of campaign polls on electoral information seeking. *Electoral Stud.* 40, 146–157.
- Sheffer, L., 2020. Partisan in-group bias before and after elections. *Electoral Stud.* 67, 102191.
- Shepherd, L., Fasoli, F., Pereira, A., Branscombe, N.R., 2018. The role of threat, emotions, and prejudice in promoting collective action against immigrant groups. *Eur. J. Soc. Psychol.* 48 (4), 447–459.
- Singh, S.P., Karakoç, E., Blais, A., 2012. Differentiating winners: how elections affect satisfaction with democracy. *Electoral Stud.* 31 (1), 201–211. <https://doi.org/10.1016/j.electstud.2011.11.001>
- Singh, S.P., Thornton, J.R., 2019. Elections activate partisanship across countries. *Am. Polit. Sci. Rev.* 113, 248–253.
- Singh, S.P., Thornton, J.R., 2024. Does the salience of partisan competition increase affective polarization in the United States? *Polit. Res. Q.* 77 (1), 45–58.
- Slothuus, R., de Vreese, C.H., 2010. Political parties, motivated reasoning, and issue framing effects. *J. Polit.* 72 (3), 630–645. <https://doi.org/10.1017/S002238161000006X>
- Slothuus, R., Bisgaard, M., 2021. How political parties shape public opinion in the real world. *Am. J. Polit. Sci.* 65 (4), 896–911. <https://doi.org/10.1111/ajps.12550>
- Smurda, J.D., Wittig, M.A., Gokalp, G., 2006. Effects of threat to a valued social identity on implicit self-esteem and discrimination. *Group Process. Intergroup Relat.* 9 (2), 181–197.
- Tajfel, H., Turner, J.C., 1979. An integrative theory of intergroup conflict. In: Austin, W.G., Worchel, S. (Eds.), *The Social Psychology of Intergroup Relations*. Brooks/Cole, Monterey, CA, pp. 33–37.
- Trumm, S., Sudulich, L., Townsley, J., 2017. Information effect on voter turnout: how campaign spending mobilises voters. *Acta Polit.* 52.
- Turner, J.C., Hogg, M.A., Oakes, P.J., Reicher, S.D., Wetherell, M.S., 1987. *Rediscovering the Social Group: A Self-Categorization Theory*. Blackwell.
- Wagner, M., 2021. Affective polarization in multiparty systems. *Electoral Stud.* 69.
- Webster, S.W., Abramowitz, A.I., 2017. The ideological foundations of affective polarization in the U.S. Electorate. *Am. Polit. Res.* 45 (4), 621–647.
- Whiteley, P.F., Seyd, P., 1994. Local party campaigning and electoral mobilization in Britain. *J. Polit.* 56 (1), 242–252. <https://doi.org/10.2307/2132356>
- Whiteley, P.F., Seyd, P., 2003. Party election campaigning in Britain: the Labour Party. *Party Polit.* 9 (5), 637–652. <https://doi.org/10.1177/13540688030095007>